CONVOLUTIONAL PHOTONIC RESERVOIR FOR FINANCIAL SWAPTION SURFACE FORECASTING

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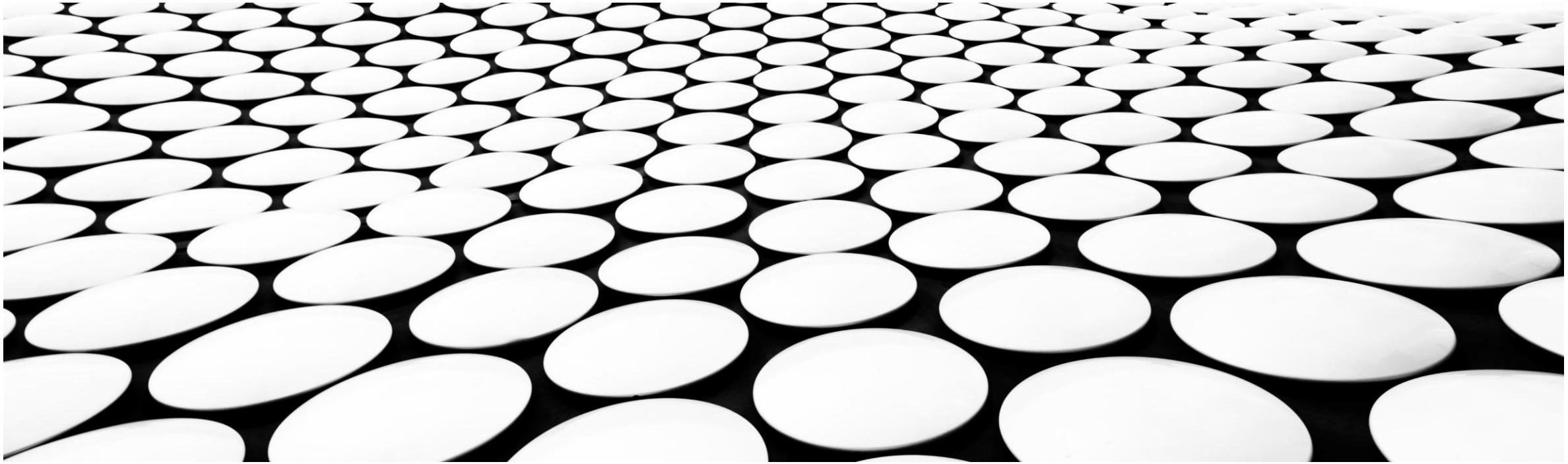
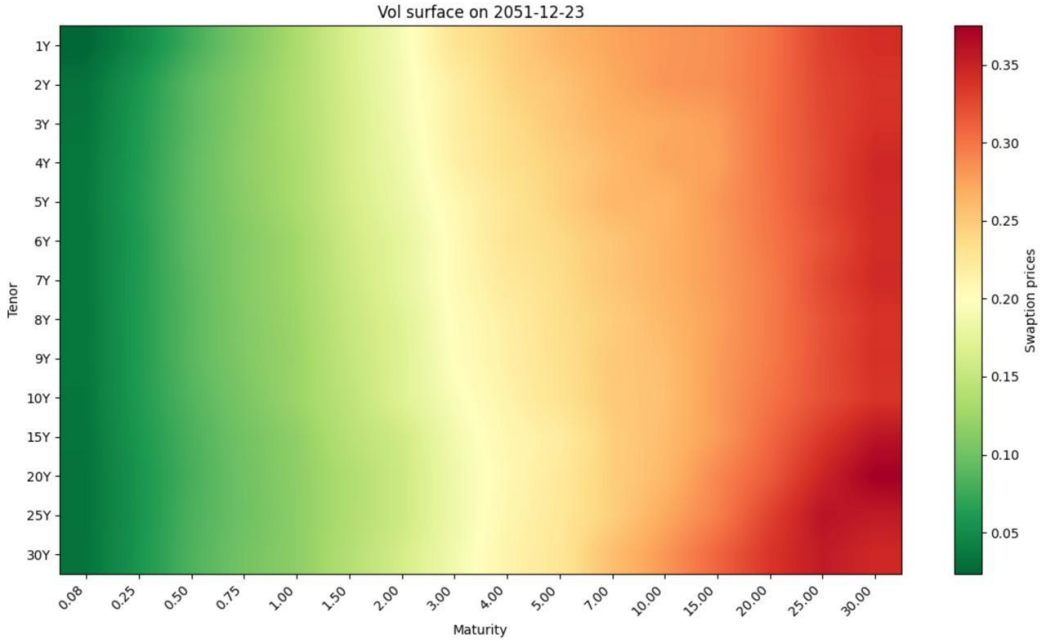
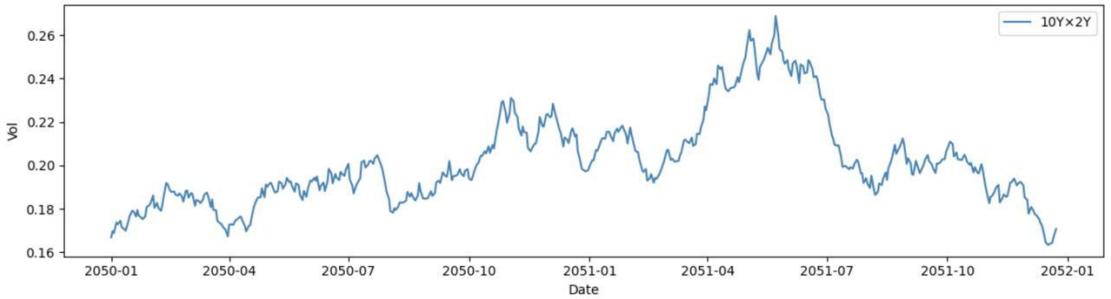


SWAPTION SURFACE FORECASTING

- A **swaption** is an option granting the right to enter an interest rate swap at a future date.
- Swaption surface forecasting predicts how implied volatilities across option maturities and swap tenors will evolve. These forecasts support more accurate pricing, risk management, and strategy development by anticipating shifts in interestrate volatility.





CLASSICAL VS QUANTUM RESERVOIRS

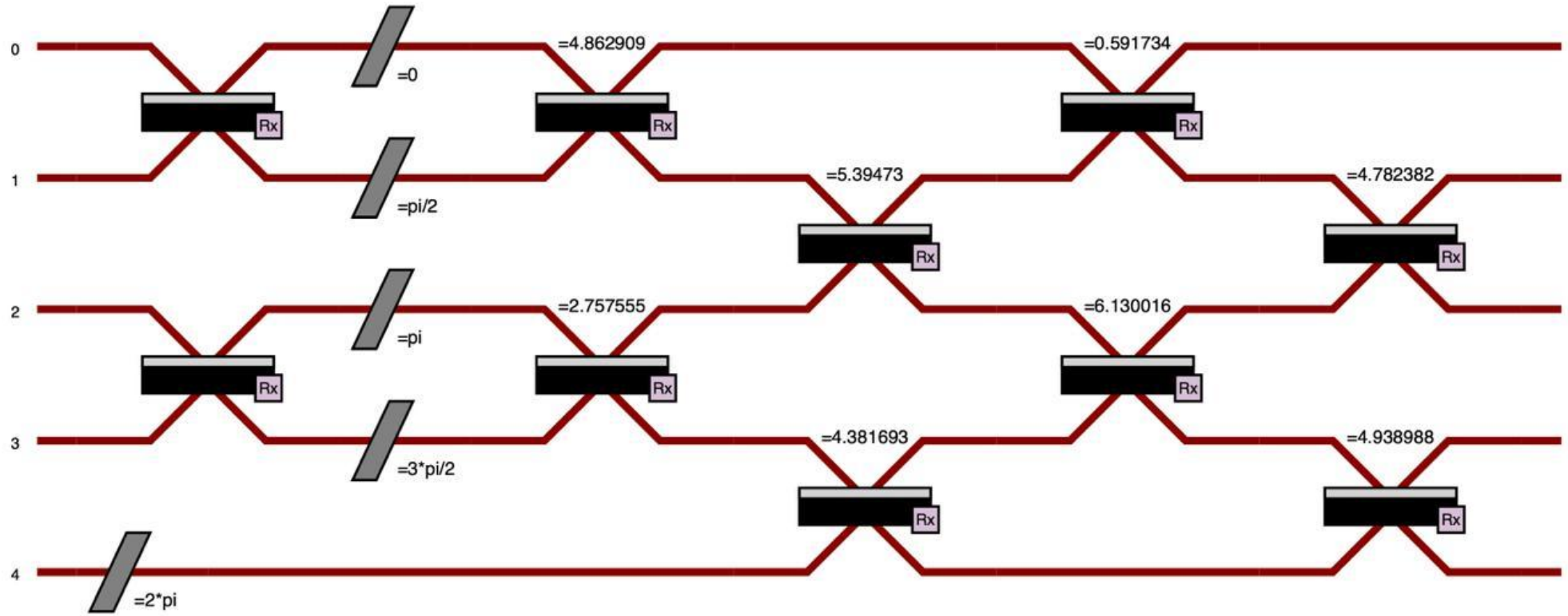
- A classical reservoir is a fixed, high-dimensional dynamical system, often a **recurrent neural network**, used to transform input signals into rich internal states. Reservoir computing trains only a simple readout layer, making it fast and data-efficient while still capturing complex temporal relationships. It is well-suited for **forecasting**, classification, and control tasks involving **nonlinear, time-dependent patterns**

- Quantum reservoir computing leverages superposition and entanglement to **achieve rich nonlinear transformations** with **minimal** training. This enables efficient learning for temporal, sequential, and pattern recognition tasks. It offers the potential for speedups, lower training cost, and improved performance on problems where classical models struggle.

SOLUTION PROPOSED

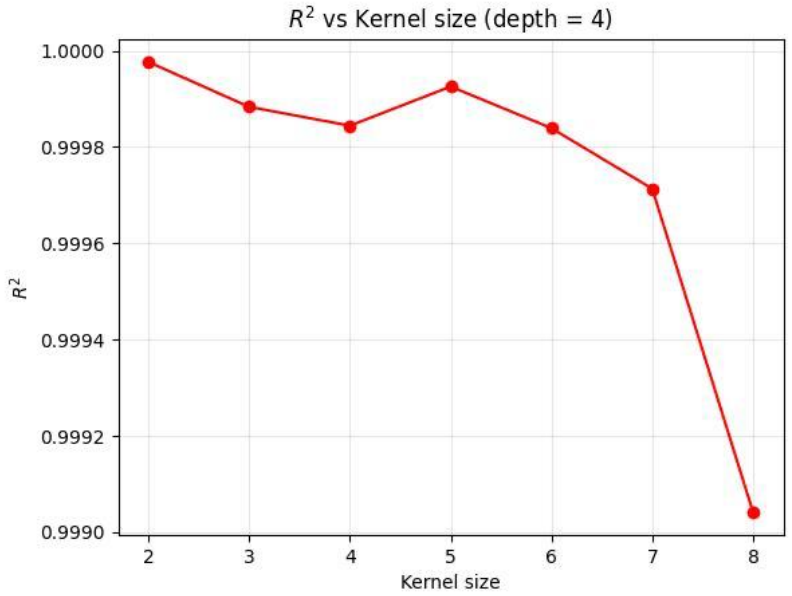
- In traditional reservoir computing, specifically using photonic circuits, the simulation of a single large system is **computationally complex** and it is hard to scale up.
- Our solution: **Convolutional Photonic Reservoir** (CPR)
- Process task into smaller chunks using a fixed sized **kernel.**
- This approach **offers efficiency and scalability**.
- Before encoding the data, a set of alternating **beam splitters** is added to distribute the input light across all modes. The scaled data is encoded into the circuit **using phase shifters**.
- The reservoir is a deep network of beam splitters with **fixed random phases** to add non-linearity into the model.
- Measurement is performed and the probability for each state is recorded and used to train the regression model (**Ridge** regression).

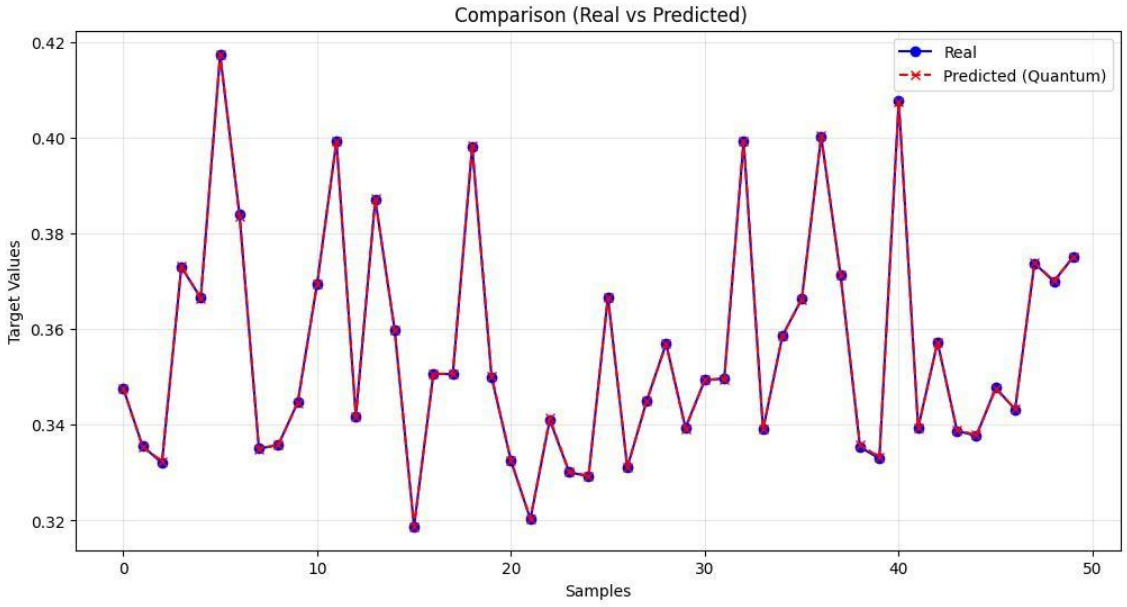
PHOTONIC CIRCUIT

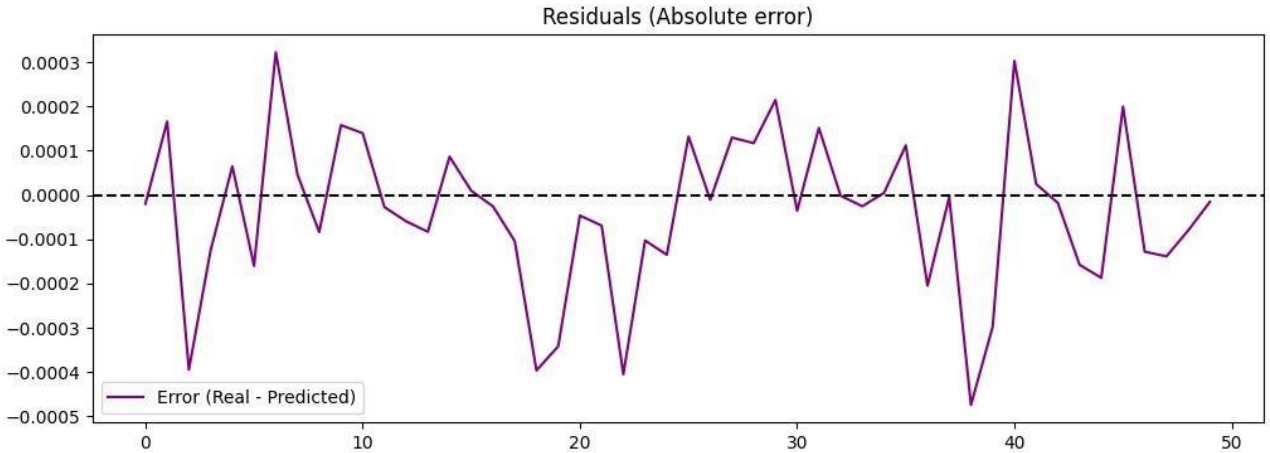


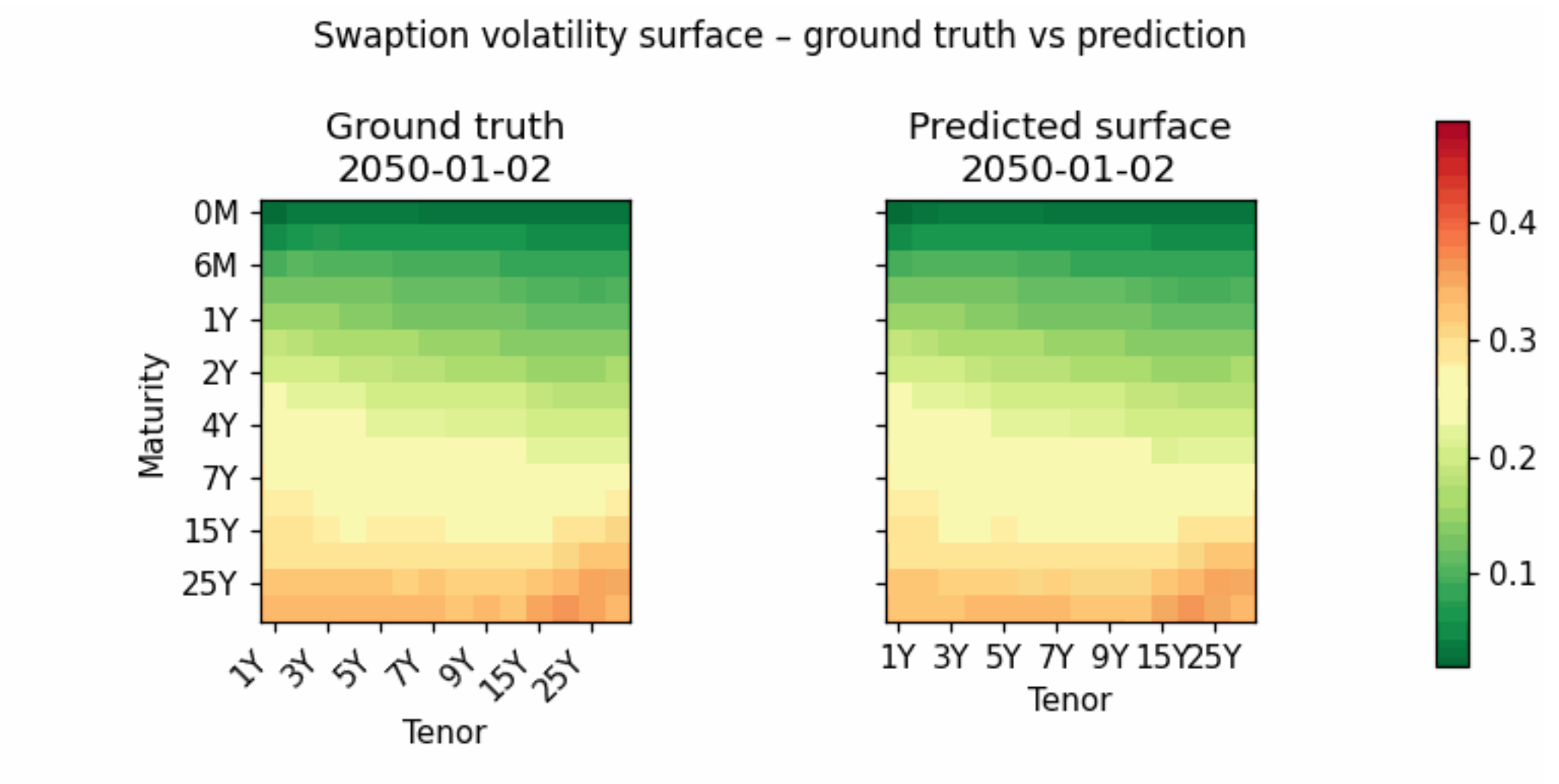
RESULTS OBTAINED

- **MSE** (Mean Squared Error): 3.9215 e10-8
- **MAE** (Mean Absolute Error): 1.4024 e10-4
- **R2 Score**: 99.993%
- **Time Execution**: 00:43 seconds

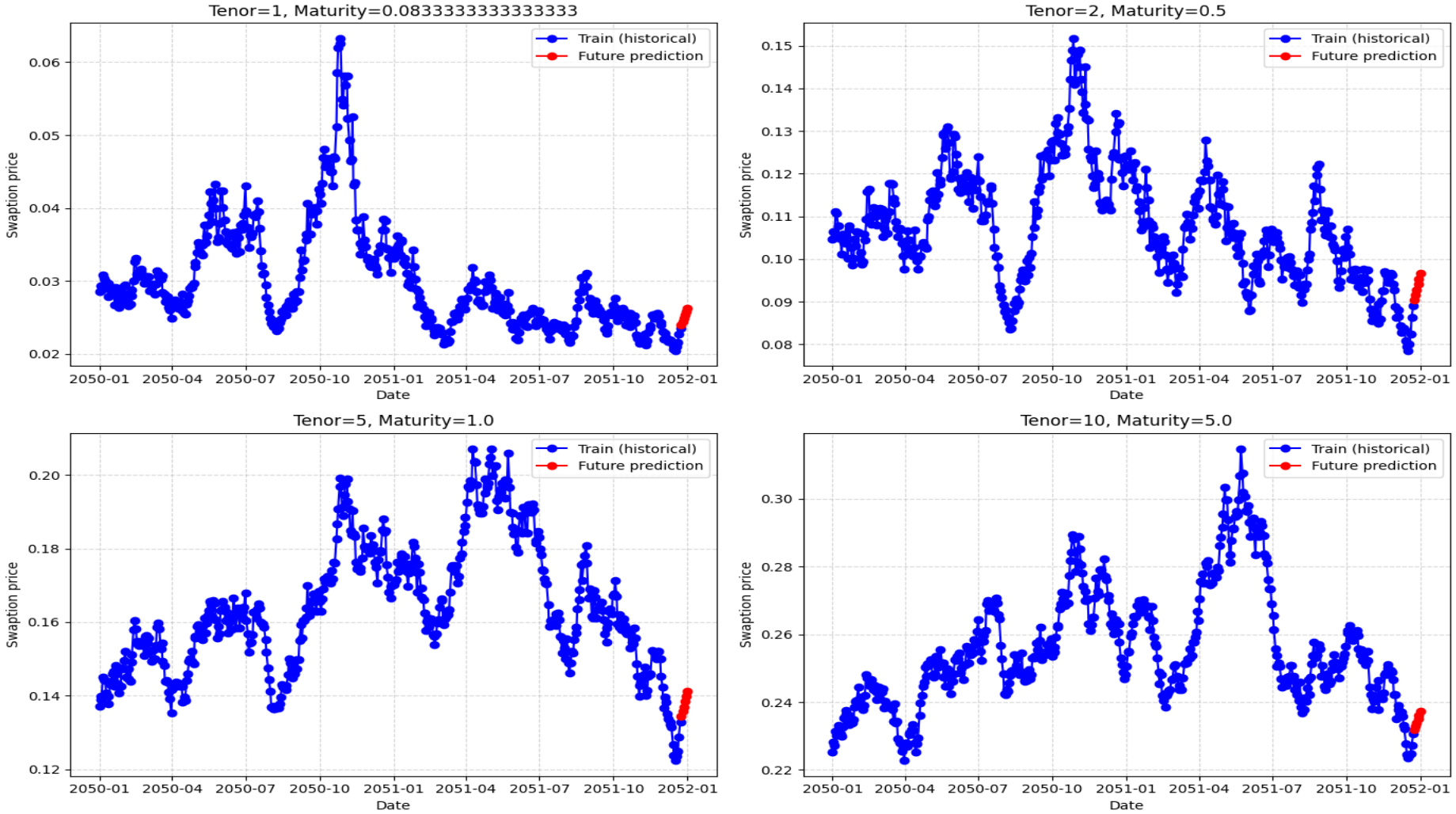






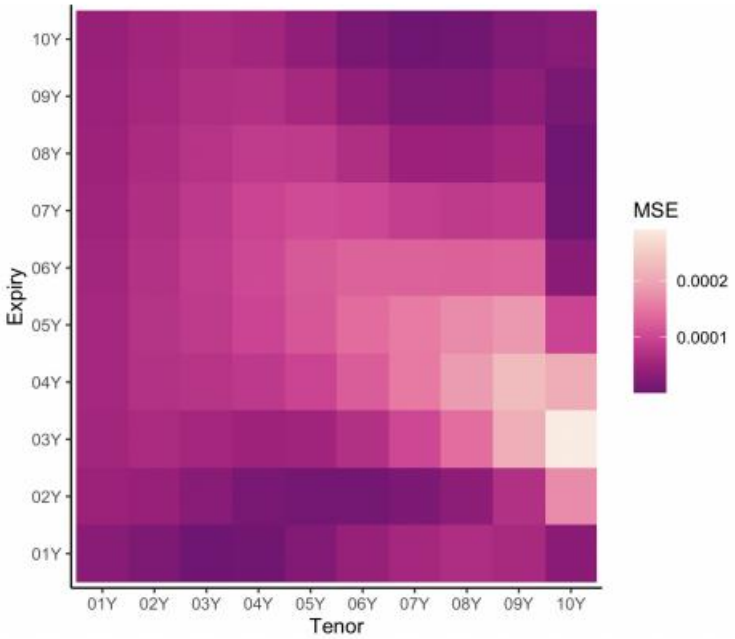

Swaption volatility surface – ground truth vs prediction
Ground truth
2050-01-02
Predicted surface
2050-01-02
Maturity
0M
6M
1Y
2Y
4Y
7Y
15Y
25Y
1Y
3Y
5Y
7Y
9Y
15Y
25Y
Tenor
0.4
0.3
0.2
0.1

PREDICTED FUTURE DATA

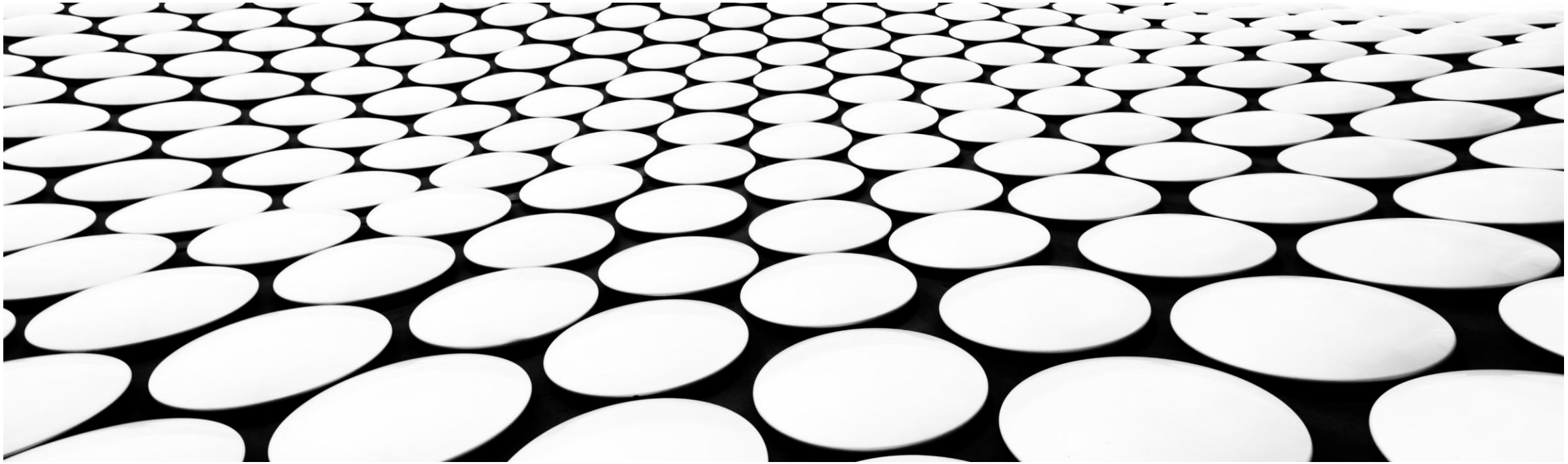


CONCLUSIONS

- The Convolutional Photonic Reservoir achieves consistently **superior** predictive accuracy, outperforming classical models such as ANNs and even advanced Quantum Reservoirs (e.g., arXiv:2505.13933), indicating a tangible **quantum advantage** for time-series forecasting.
- The model introduces a **novel kernel** approach, where the photonic reservoir acts as a high-dimensional **nonlinear** feature map, a configuration not previously explored in the literature.
- Its **hybrid** design combines classical efficiency with the expressive **power** and **parallelism** of photonic quantum dynamics, enabling state-of-the-art performance.
- Future work includes a **Modular Reservoir Design**: multiple specialized photonic reservoirs, each optimized for a specific quantum operation (e.g., entangling beam splitters or phase-shifter rotations), to enhance scalability and expressiveness.



[http://hdl.handle.net/10230/56573]

THANKS FOR THE ATTENTION!

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